

PANISARA THEPTHONG

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PERSONAL PROFILE

I am a PhD student in Space Physics at LPC2E, focusing on solar wind data analysis using spacecraft measurements. My research interests center on solar phenomena that drive and modulate interplanetary space weather, investigated through a combination of in-situ and remote-sensing observations. I am currently working on Type III radio burst analysis using data from Parker Solar Probe (PSP) and Solar Orbiter (SolO). I am skilled in scientific data analysis, in-situ measurements, and quantitative research.

EDUCATION

LPC2E, University d'Orléans

Orléans, France

PhD

October 2024 - Present

Thesis Project: Analyzing Solar Type III Radio Bursts with Solar Orbiter and Parker Solar Probe

Supervisor: Matthieu Kretzschmar and Milan Maksimovic

Kasetsart University

Bangkok, Thailand

Master of Science (M.Sc.) in Physics

June 2022 - July 2024

GPA. 3.83

Thesis Project: Scale and Time Dependence of Alfvénicity in the Solar Wind as Observed by the Parker Solar Probe.

Supervisor: Asst. Prof. Peera Pongkitiwanichakul and Prof. David Ruffolo

Bachelor of Science (B.Sc.) in Physics

August 2018 - April 2022

GPA. 3.55 (First-class honors)

The Prince Royal's College

Chiangmai, Thailand

School

2005 - 2018

TECHNICAL EXPERTISE AND LANGUAGES

Programming

Python, Fortran, Javascript

Languages

Thai (Native), English (Professional working proficiency), French (Beginner)

PUBLICATIONS

1. Panisara Thepthong et al. "Scale and Time Dependence of Alfvénicity in the Solar Wind as Observed by the Parker Solar Probe". In: 962.1, 37 (Feb. 2024), p. 37. DOI: 10.3847/1538-4357/ad1592. arXiv: 2312.08707 [astro-ph.SR]
2. Joshua Goodwill et al. "Parker Solar Probe analysis across the Alfvénic transition: velocity shear, magnetic deflection, and switchback formation". In: 547.2, stag242 (Apr. 2026), stag242. DOI: 10.1093/mnras/stag242
3. Subash Adhikari et al. "Characterization of the Alfvén Transition in the Young Solar Wind using Parker Solar Probe Observations Approaching Solar Maximum". In: 997.2, 259 (Feb. 2026), p. 259. DOI: 10.3847/1538-4357/ae2c78
4. David Ruffolo et al. "Observed Fluctuation Enhancement and Departure from WKB Theory in Sub-Alfvénic Solar Wind". In: 977.1, L19 (Dec. 2024), p. L19. DOI: 10.3847/2041-8213/ad9727. arXiv: 2409.02612 [astro-ph.SR]
5. Rohit Chhiber et al. "The Alfvén transition zone observed by the Parker Solar Probe in young solar wind - global properties and model comparisons". In: 533.1 (Sept. 2024), pp. L70–L75. DOI: 10.1093/mnras/1/slae051. arXiv: 2405.10437 [astro-ph.SR]

6. David Ruffolo et al. "Domains of Magnetic Pressure Balance in Parker Solar Probe Observations of the Solar Wind". In: 923.2, 158 (Dec. 2021), p. 158. DOI: 10.3847/1538-4357/ac2ee3. arXiv: 2110.08506 [astro-ph.SR]

CONFERENCES AND WORKSHOPS

1. **2nd European solar physics division (ESPD) Summer School** Dubrovnik, Croatia
Attendee April - May 2026
2. **Solar Orbiter - IRIS - ADITYA - Joint Workshop** Berlin, Germany
Poster Presentation March 2026
Presented a poster titled "Investigating Fluctuations in Solar Type III Radio Burst Spectra: Observations with Parker Solar Probe and Solar Orbiter"
3. **RPW Consortium 32** Kiruna, Sweden
Oral Presentation January 2026
Presented an oral talk titled "What Can We Learn From Type III Radio Burst Fluctuations?"
4. **Planetary, Solar and Heliospheric Radio Emissions X (PRE)** Marseille, France
Oral Presentation Jun 2025
Presented an oral talk titled "Characterization of Fluctuations in Solar Type III Radio Spectra Observed by PSP and SolO."
5. **School on Cross-scale Coupling of Heliophysics Systems** L'Aquila, Italy
Attendee May 2025
6. **European Geosciences Union (EGU) General Assembly 2025** Vienna, Austria
Poster Presentation May 2025
Presented a poster titled "Joint Analysis of Solar Type III Radio Bursts with Parker Solar Probe and Solar Orbiter" and attended sessions aligned with my research interests.
7. **Journées Jeunes Chercheurs (JC1)** Toulouse, France
Attendee Apr 2025
Met CNES correspondents and participated in scientific discussions with early-career researchers.
8. **RPW Consortium 31** Athens, Greece
Oral Presentation Nov 2024
Gave a presentation on Alfvénicity in the solar wind (previous work) and preliminary results from Solar Type III radio-burst analysis using Parker Solar Probe and Solar Orbiter observations.
9. **Particle Detection: From Ground to Space and Space Weather Impacts (Workshop)** Chiang Mai, Thailand
Attendee Feb 2024
Participated in talks and hands-on activities focused on space weather and particle detection.
10. **The 14th International Fundamental Science Congress (IFSC)** Bangkok, Thailand
Oral Presentation Aug 2023
Presented an oral talk titled "Scale and Time Dependence of Alfvénicity in the Solar Wind as Observed by the Parker Solar Probe."
11. **Thai Space Physics 2023** Bangkok, Thailand
Oral Presentation Aug 2023
Presented results on the scale and time dependence of Alfvénicity in the solar wind as observed by the Parker Solar Probe.
12. **American Geophysical Union (AGU) Fall Meeting 2022** Online
Poster Presentation Dec 2022
Presented a poster titled "Scale-Dependent Alfvénicity of Super-Alfvénic and Sub-Alfvénic Solar Wind Observed during Parker Solar Probe's 8th Orbit."
13. **Second Annual Parker Solar Probe Conference (Parker Two)** Online
Poster Presentation Jun 2022

Presented a poster titled “Alfvénicity of Velocity and Magnetic-Field Increments during Parker Solar Probe 8th Orbit.”

14. **European Geosciences Union (EGU) General Assembly 2022** Online
Short Oral Presentation May 2022
Presented a short oral talk titled “Alfvénicity of Velocity and Magnetic-Field Increments Observed by Parker Solar Probe.”
15. **Thai Space Physics 2022** Bangkok, Thailand
Oral Presentation Sep 2022
Presented results on magnetic-field domain structure and progress on Alfvénicity in the solar wind.
16. **Thai Space Physics 2021** Bangkok, Thailand
Attendee Aug 2021
Attended research presentations on space plasma physics and cosmic rays from multiple universities in Thailand.

EXPERIENCE

Teaching Experience

- Private Mathematics Tutor 2022 - 2023
- Teaching Assistant, Kasetsart University 2021
 - General Physics I
- Tutor for Calculus I 2019
 - Volunteered as a peer tutor for first-year students in the department’s academic support activities.

Outreach Activities

- Staff in Astronomy Club, Kasetsart University 2018 - 2022
 - Organized the camp and activities to provide knowledge in Astronomy to undergraduate students.
- Physics Battle 8th 2020
 - Organized the Physics competition for high school students and handled administrative paperwork.

Personal Projects

- Role & Roll Dice Roller 2025 - Present
 - Developed a web-based dice-rolling tool with character sheet support for a Thai tabletop role-playing game (TTRPG) system. (https://look2ed.github.io/rolenroll_interactive_sheet/)

SCHOLARSHIPS AND AWARDS

- Doctoral Contract (PhD Grant), LPC2E (CNRS/Université d’Orléans) 2024–2027
- Outstanding Academic Achievement Award (M.Sc. graduate), Academic Year 2023 2024
- Best Oral Presentation Award (Physical Science Session), 14th IFSC 2023
- Kasetsart University Graduate School Fellowship (support for M.Sc. studies) 2022–2023